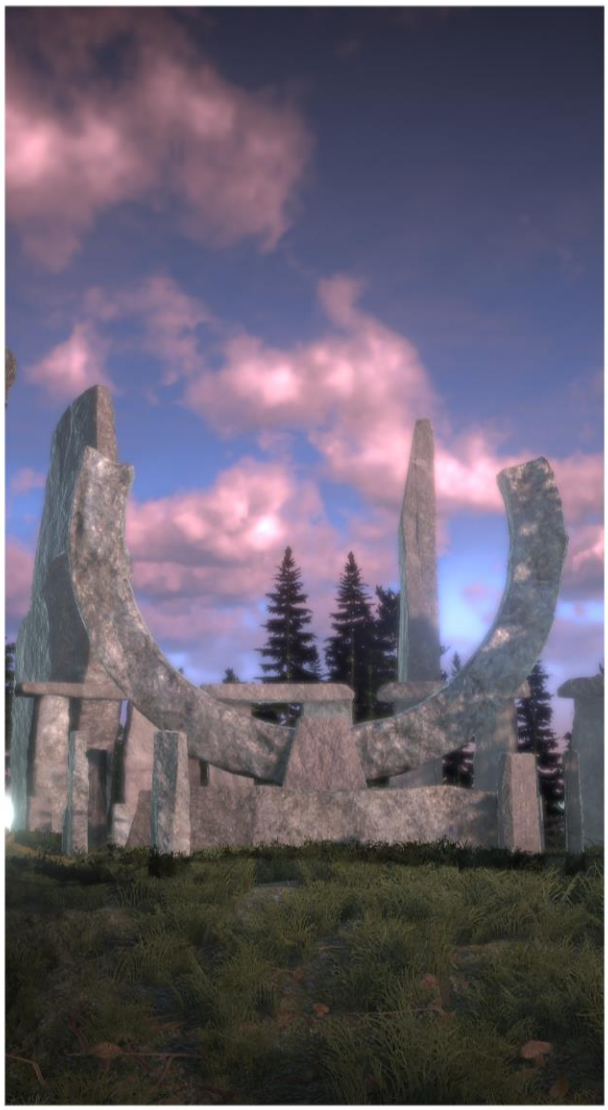


Stone Circle Generator

GENERATE ENDLESS UNIQUE STONE CIRCLES

Luke Higgs | VarcanRex | © 7/8/2022

About this Tool

This tool is designed as an easy solution to quickly generate stone circles like what you see across the UK, Ireland, and other countries. To achieve a "natural" look to stone placements the tool relies on randomization options. These settings can be disabled to achieve even and oriented circles if that is what you need.

I have included a large library of realistic stones modeled after common stones. I have also included several altar, portal, and spires objects for you to use also. There are also sample presets and prefabs you can use out of the box or as starting points. These are located in "Assets/VarcanRex/Stone_Circle_Generator /UserData/SavedPrefabs" and "Assets/VarcanRex/Stone_Circle_Generator /UserData/SavedPresets" directories.

You are not limited by the included libraries. You may easily add new folders to the libraries and add your own prefabs to use. In the following pages I'll go through each setting in the Manager and Ring components.

Contact & Support

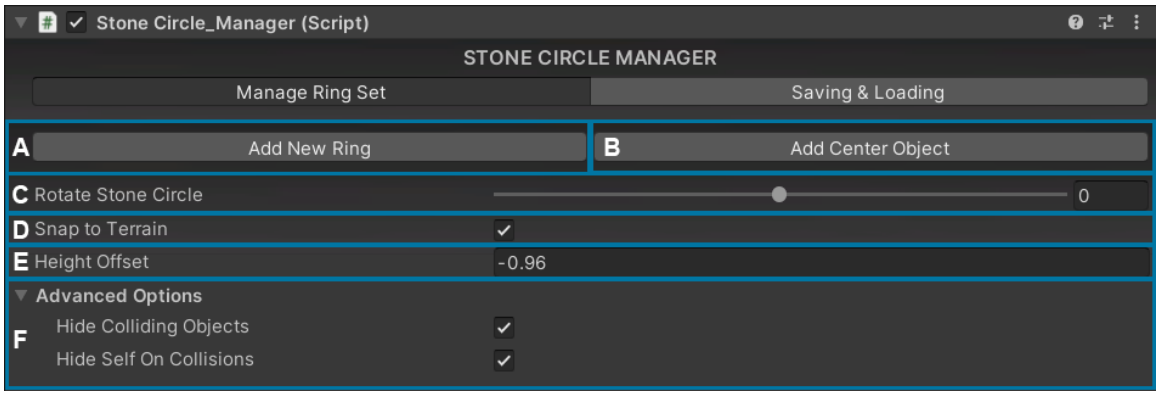
For any questions, please send an email to varcan.rex@outlook.com, please include "Stone Circle Generator" in the subject line.

Features Overview

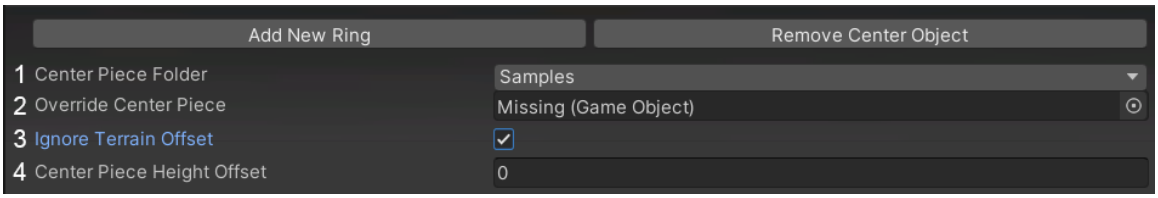
To add a Stone Circle to your scene go to *Tools->Stone Circle Generator->New Stone Circle*. A new Stone Circle Manager will be added to your scene and placed in front of the scene camera. Alternately you can create a new Empty and add the StoneCircle_Manager.cs component.

Tip: If you have a Terrain active in your scene you will have the option to snap endpoints to the terrain. Use this together with the height offset to adjust your endpoints and ensure they appear "buried" in the ground.

Stone Circle Manager – Managing Ring Sets



A. Add a new ring set, there is no limit to the number of rings you can add.
B. Add a Center Object:

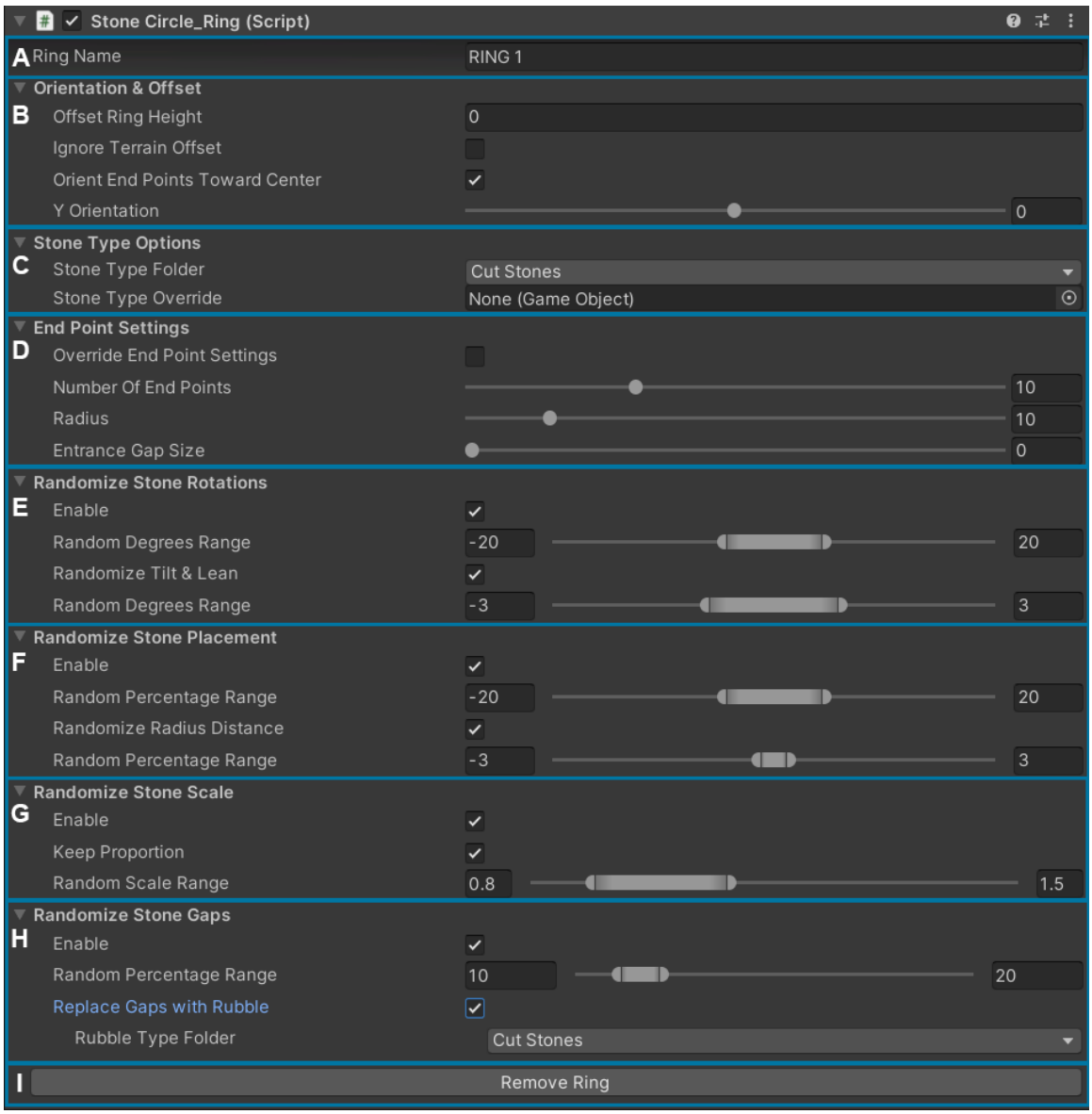


   1. The Center Piece Folder drop down allows you to select a random prefab from a folder in the …/UserData/CenterTypes directory. You can add additional folders with your own custom prefabs (see the Using Custom Prefabs section).
   2. Use the "Override Center Piece" option to drag and drop a prefab to use.
   3. You can check "Ignore Terrain Offset" if you don't want the "Snap to Terrain" feature to affect the height offset.
   4. "Center Piece Height Offset" allows you to raise/lower the height of your center piece independent of other offset settings.

C. Rotates the entire circle on the Y axis.
D. Snap each end point and center piece to the terrain.
E. Adjust the height of the entire circle.
F. Hide objects that endpoints and center pieces collide with, alternately you can check "Hide Self on Collisions" and the end point or center piece will be hidden instead.

Stone Circle Manager – Ring Settings

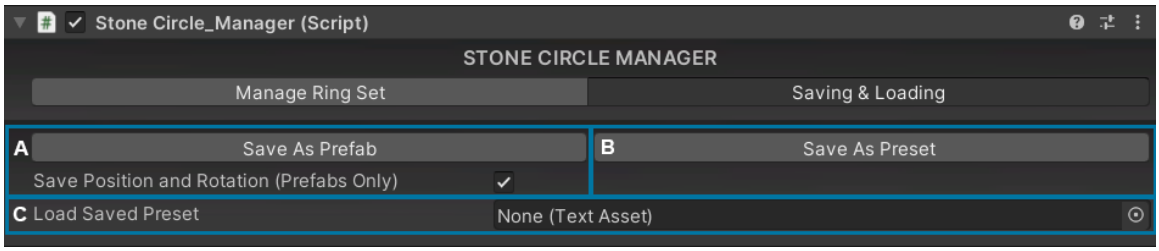
NOTE: Most of the settings in the Stone Circle Generator use ranges of randomness to generate unique and "natural" configurations. This results in the stones "jittering" when making adjustments. This is normal and due to random selections made within the ranges whenever an adjustment is made.



A. Set a label for each ring.
B. **Orientation & Offset.** Adjust the Y (Height) offset for each ring, check "Ignore Terrain Offset" if you don't want this ring to be affected by "Snap to Terrain" adjustments. Use "Orient End Points Toward Center" if you want the end points to face the center. Use "Y Orientation" to rotate the ring on the Y axis.

C. **Stone Type Options.** Select a folder from the "…/UserData/EndPointTypes" directory. Stones will be randomly selected from the prefabs in the selected folder. Create additional folders in this directory with your own prefabs (See the section on using custom prefabs). Use the Stone Type Override to drag and drop a single prefab to be used as end points.
D. **End Point Settings.** Use "Override End Point Settings" to manually enter a radius and number of end points, otherwise use the sliders. End point overrides are capped at 1000 end points. Adjusting the "Entrance Gap Size" will open up the circle (hiding endpoints). This setting is a percentage value of end points.
E. **Randomize Stone Rotations.** Each stone (end point) will be rotated a random degree between the range sliders you select. End points will also be randomly flipped 180 degrees. Tilt and Lean will randomly rotate the end point on the X & Z axis to simulate stones leaning over time.
F. **Randomize Stone Placement.** The first setting will randomly adjust how close each end point is to the other end points within the degree range you select. "Randomize Radius Distance" with adjust the distance from the center of each end point. This setting uses a percentage range of the original distance from the center.
G. **Randomize Stone Scale.** Use the range slider to randomly adjust the scaling of each end point. Select "Keep Proportion" to scale evenly on all axes.
H. **Randomize Stone Gaps.** Use these settings to randomly remove end points and create gaps. Optionally you can use "Replace Gaps with Rubble" to replace the gaps with random prefabs from the selected folder.
I. **Remove Ring.** This removes the ring and cannot be un-done.

Stone Circle Manager – Saving and Loading



A. Save the current ring set as a prefab (prefabs are saved in the "…/UserData/SavedPrefabs" folder). Check "Save Position and Rotation (Prefabs Only)" if you want to convert your ring set to a prefab to be used on a terrain and you want it to have the exact location.
B. Save current ring set as a preset (presets are saved in the "…/UserData/SavedPresets" folder). Presets are JSON files and can be edited with a text editor if needed.
C. Drag and drop a saved preset here to load presets.

User Data Location & Using Custom Prefabs

All user data is stored in the "Assets/VarcanRex/Stone_Circle_Generator/UserData" root directory. This directory shouldn't be renamed or removed.

Create folders inside the "…/UserData/EndPointTypes" and place your prefabs in the folder, the Stone Ring Manager will automatically pick up the new folders as long as they have prefabs in them. Center Pieces should be added to the "…/UserData/CenterTypes" directory.

NOTE: It's best to add your own colliders to your custom prefabs. Stone Ring Generator will add colliders if none are found but they may not be exactly as you want them. If a prefab doesn't require a collider (a particle emitter for example) place a small sphere collider and set it as trigger.

That's it!

You can use this tool out-of-the-box to generate endless stone circle combinations, however I encourage you to use your own prefabs and get creative. This simple tool can be used to make all kinds of structures that are based on concentric rings, think the city of Atlantis!

URP & HDRP: Materials can be easily converted to URP or HDRP by using the standard Unity conversion methods (linked). If you have any trouble, don't hesitate to contact us.

Limitations

Hiding endpoints & objects that have LOD groups do not work properly as the script uses colliders to detect colliding meshes. If you zoom the scene camera close enough so that LOD0 is displayed the collisions will work properly, you can then manually remove the colliding objects.